

TORUP, Sweden

WMO: 026180

Lat: 56.9494N

Lon: 13.0601E

Elev: 427

StdP: 14.47

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.9	10.5	1.5	6.0	6.8	6.6	7.8	14.2	25.4	40.9	21.0	40.3	2.4	20	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	80.3	63.4	77.1	62.1	73.9	60.5	66.5	74.5	64.6	72.2	62.9	70.3	6.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	90.1	68.4	62.0	84.4	67.0	60.2	79.0	65.2	31.3	74.9	29.9	72.0	28.7	70.4	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	15.6	13.6	DB	-0.6	85.4	7.4	4.2	-5.9	88.4	-10.2	90.8	-14.4	93.2	-19.7	96.2	(4)
(5)			WB	-1.0	69.6	7.3	2.2	-6.3	71.2	-10.6	72.5	-14.7	73.7	-20.0	75.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	44.9	29.4	30.1	34.4	43.3	51.3	57.0	61.7	60.8	53.7	45.5	38.2	32.8	(6)
(7)		DBStd	13.18	8.29	7.58	6.97	6.46	6.41	5.03	4.99	4.57	5.22	6.37	6.89	8.85	(7)
(8)		HDD50	3038	638	557	483	218	59	4	0	0	26	164	355	534	(8)
(9)		HDD65	7375	1103	977	948	651	428	245	132	145	338	605	804	999	(9)
(10)		CDD50	1191	0	0	0	18	98	215	361	336	138	24	1	0	(10)
(11)		CDD65	52	0	0	0	0	2	6	28	16	0	0	0	0	(11)
(12)		CDH74	676	0	0	0	7	37	139	293	191	9	0	0	0	(12)
(13)		CDH80	97	0	0	0	0	1	21	48	27	0	0	0	0	(13)
(14)	Wind	WSAvg	6.1	6.3	6.5	6.6	6.2	6.0	5.9	5.7	5.3	5.5	6.1	6.4	6.5	(14)
(15)	Precipitation	PrecAvg	45.2	4.3	3.2	2.9	2.3	2.7	3.7	4.2	4.6	4.2	4.6	4.3	4.3	(15)
(16)		PrecMax	55.6	9.1	6.9	7.6	5.0	5.0	6.7	8.9	7.4	9.5	10.6	7.0	9.8	(16)
(17)		PrecMin	32.4	0.3	0.6	1.1	0.3	0.4	0.1	0.6	0.9	1.1	1.3	2.2	0.7	(17)
(18)		PrecStd	6.8	2.4	1.8	1.5	1.3	1.1	1.7	2.1	1.8	2.0	2.3	1.4	2.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	47.5	56.6	71.9	78.1	82.9	84.9	83.2	75.3	63.0	53.6	49.4	(19)
(20)			MCWB	44.4	42.2	45.8	55.3	59.8	63.7	65.9	64.2	61.2	55.8	52.3	48.5	(20)
(21)		2%	DB	42.9	44.1	51.1	65.6	74.1	78.1	80.4	79.2	69.7	58.2	51.3	46.8	(21)
(22)			MCWB	41.8	41.8	43.3	51.4	57.6	61.0	64.6	63.4	59.1	54.9	49.7	45.8	(22)
(23)		5%	DB	41.2	41.8	46.7	61.0	70.1	73.7	77.2	75.2	66.0	56.1	49.4	45.1	(23)
(24)			MCWB	40.0	40.1	41.5	49.2	55.5	59.1	63.4	62.1	58.1	53.5	48.0	44.2	(24)
(25)		10%	DB	39.6	39.8	43.9	56.3	65.5	69.4	73.5	71.5	62.8	54.4	47.5	43.4	(25)
(26)			MCWB	38.6	38.2	40.2	46.7	53.9	57.8	61.2	61.1	56.8	52.0	46.3	42.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.7	44.8	47.7	57.2	63.6	66.3	69.4	68.5	64.2	58.2	52.6	48.8	(27)
(28)			MCDWB	45.0	46.1	52.9	69.3	73.4	78.2	78.3	75.6	70.3	60.2	53.3	49.2	(28)
(29)		2%	WB	42.2	42.4	44.9	53.2	60.3	63.4	67.2	66.0	61.2	55.7	50.1	46.0	(29)
(30)			MCDWB	42.6	43.7	48.8	62.5	69.2	73.1	75.4	72.8	66.2	57.4	51.0	46.5	(30)
(31)		5%	WB	40.3	40.4	42.9	50.5	57.6	61.2	65.2	64.1	59.4	54.1	48.2	44.3	(31)
(32)			MCDWB	41.0	41.5	45.7	58.2	65.8	70.2	73.1	70.4	63.7	55.8	49.2	44.9	(32)
(33)		10%	WB	38.8	38.5	41.0	48.0	55.0	59.2	63.3	62.5	57.9	52.5	46.5	42.6	(33)
(34)			MCDWB	39.6	39.5	43.3	54.1	63.6	66.4	71.0	69.2	61.7	54.2	47.4	43.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.5	10.1	14.2	18.9	19.6	18.4	17.5	17.4	16.2	11.9	8.4	8.0	(35)
(36)			MCDBR	7.2	11.0	19.1	28.1	28.8	29.2	25.2	25.4	21.9	13.1	8.9	7.5	(36)
(37)			MCWBR	7.2	8.9	12.2	15.3	14.7	14.4	11.6	11.7	12.8	9.8	8.4	7.6	(37)
(38)		5% WB	MCDWB	7.2	8.4	15.3	24.3	22.9	22.9	20.4	19.0	18.0	11.7	8.4	7.4	(38)
(39)			MCWBR	7.3	7.5	11.2	14.2	12.8	12.9	10.8	10.7	12.0	9.5	8.0	7.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.298	0.326	0.345	0.381	0.378	0.369	0.391	0.391	0.359	0.347	0.323	0.288	(40)	
(41)		taud	2.266	2.308	2.343	2.306	2.377	2.426	2.379	2.384	2.458	2.408	2.326	2.229	(41)	
(42)		Ebn at Noon	183	229	257	264	273	277	267	258	251	220	172	151	(42)	
(43)		Edh at Noon	16	23	29	35	35	34	35	32	26	21	16	13	(43)	
(44)	All-Sky Solar Radiation	RadAvg	125	316	753	1241	1623	1726	1593	1270	889	420	157	86	(44)	
(45)		RadStd	24	50	76	170	154	137	200	118	107	67	29	10	(45)	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (70 neighbors)	+0.79	N/A	N/A	N/A	N/A	-227	-320	N/A	N/A	(47)

Nomenclature: See separate page